Koblitz cover transfers: Round24

Explicit infinity fiber, original valuations and constant field

2026-10-28 UTC. Exact computations and written geometric derivations; no IC cost, relation-rank or official promotion claim.

A complete proper fiber from original source charts

For the unchanged degree16 incidence field over $E : v^2 + v = u^3 + u$, the fiber above infinity consists of four rational points, with ramification indices1,1,7,7. In the actual base chart $t = \frac{u}{v}$, $q = \frac{1}{t^3 v}$, the equation is $q + 1 + (1 + t)t^3 q^2 = 0$ and O has t=0,q=1.

The degree8 intermediate tower is $H(Z) = Z^8 + t^{11} dZ^2 + A^2 t^{11} dZ + t^{14} e$, with d,e,A units of residue1, followed by $\lambda^2 + A\lambda = Z + b$. The unramified chart has $Z = t^3 y$. The tame chart has $t = w^7 x^2$, $Z = w^{11} x^3$, with exact inverse $w = \frac{Z^2}{t^3}$, $x = \frac{t^{11}}{Z^7}$. Native checks certify the original Tate/base substitutions, both tower identities and four unit Jacobian residues. Smooth local rings identify the normalization at these points; the proper degree sum1+1+7+7=16 excludes extra branches. The scheme/DVR argument is written in CM.md.

Actual coordinate and denominator orders

Point	e	lambda	nu	S	N	nu numerator / denominator
O1,0	1	5	0	0	0	11 / 11
O1,1	1	0	0	0	0	11 / 11
O7,0	7	11	32	44	66	109 / 77
O7,1	7	0	21	44	66	98 / 77

All residue degrees are1; b,c have orders3e,2e and A is a unit. The eight actual registered polynomial-entry orders are (3,2,2,2,5,2,0,0) on either unramified point, and (21,14,14,14,35,14,44,66) on either ramified point. S,N are the unchanged nn65,nn75 entries, with original circuit denominators1. The local zero-divisor contributions above O are16 for lambda,53 for nu, 88 for S and132 for N. Global zero/pole completeness remains open.

O24-series-02 uses the certified unique chart lifts modulo w192, checks the original nu coefficient and all five source residuals modulo w100, and displays nonzero leading coefficients below100. Denominator division leaves at least115 correct coefficients on tame branches; printed high nu tails beyond that effective precision are not claimed as true expansions.

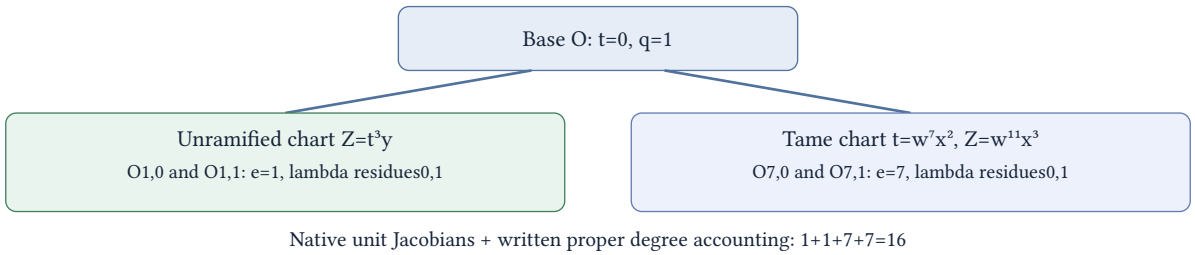
The constant field is F2

The unchanged primitive plane equation stays irreducible of lambda degree32 over F4, with exact product, separability and field controls. The constant degree divides16 and10, hence is1 or2; nonsplitting excludes2. The rational source points give a second written proof by evaluating global constants into F2. Proper geometric normality and H0=F2 imply geometric integrality (Stacks33.10.7). This does not identify an independent modular model or evaluate a factor map.

Evidence, proof scope and remaining construction

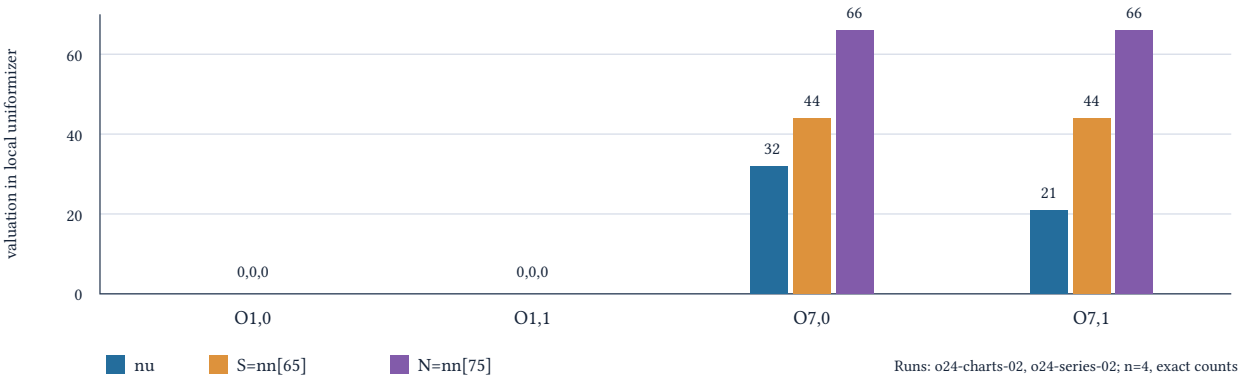
Round24: the complete source fiber above O

Actual incidence field $L/F_2(E)$, degree16; E: $v^2+v=u^3+u$. All four points have residue F_2.



Native unit Jacobians + written proper degree accounting: 1+1+7+7=16

Exact original-function orders (one normalized point per group)



Retained verification and failures

Six native calls:3 accepted positives,2 rejected implementation failures, 1 expected rejection. The corrupted actual factor product is rejected despite native exit0 and a later success marker. The failed map syntax and incorrect order helper remain in FAILURES.md with all raw output. Explicit valuation controls distinguish a unit, order1 and order7 before accepting the series.

Lean02 rebuilds all transitive project imports and audits14 new theorem declarations:0 failing,0 trusted-native. Lean01′s elaboration failure is retained separately. Exact chart identities, rational inverses and residue algebra are kernel proofs. Proper normalization, Hensel lifting, divisor interpretation and geometric constant-field identification remain written bridges, exposed in PROOF_MAP.md.

The temporary archive ignore initially excluded raw bytes from checkpoints; it is corrected, and a fresh checkpoint download confirms inclusion. No migration occurred during that defect. Final raw archive/source pins and export are separate immutable publication records. Root software and PR CI statuses are reported separately; no full green root suite is inferred.

Next discriminating construction

Resolve A0,B0,C0,D0 using source-bound finite/scaled charts; the successful O substitution is a lead to test after exact coefficient translations, not an assumed answer elsewhere. Enumerate additional zero supports by norms/source fibers. Complete divisors must precede a complete prescribed RR basis and candidate map. Acceptance still requires function-field identities, F2 descent and nonzero order11 action.

Node187 source-bound membership/complete coupled arc, full quotient saturation/proper polarized embedding and arbitrary reverse-divisor evaluation with ordinary460/367 fibers/images remain open. The n3 and same-field n5 controls and exact norm-zero recovery d=1 are preserved. No new EKS job, duplicate Diem, fifth claim or PR merge is authorized here.

Reproduction: PROTOCOL.md, protocol.json, REPLAY_NATIVE.sh, REPLAY_LEAN.sh, results.json, CM.md and source/raw manifests. The editable SVG plots recorded exact orders at four points; there is no timing or statistical comparison. Native cap:4GiB virtual memory and1200s per call.